GATE-AII BRANCHES
Engineering Mathematics



Probability and statistics

Lecture No.- 14



By- Rahul sir

Topics covered in previous lecture



Topic Expectation, variance and standard deviation

Topic Concept based on probability distribution

Topics to be Covered



Topic Uniform distribution

monday
Questions

Topic Concept based on gaussian distribution

Topic :Uniform distribution

Uniform Distribution [Continuous Prob. Distribution]

Throwing A Die : X = No. of dots X = 1, 2, 3, 4, 5, 6 (discrete Random variable)

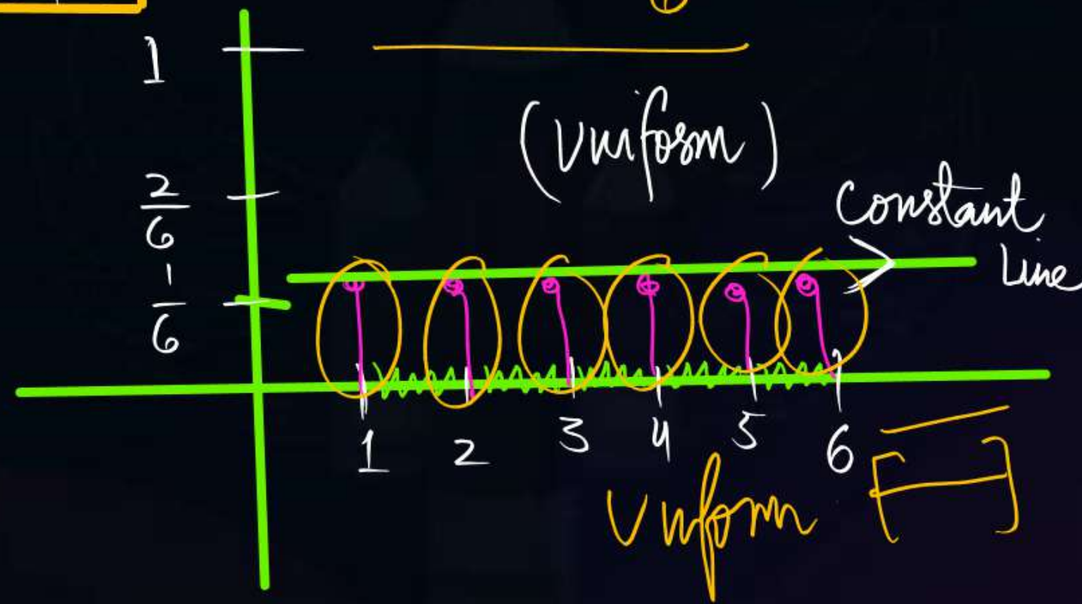
X	1	2	3	4	5	6
P(X=x)	1/6	1/6	1/6	1/6	1/6	1/6

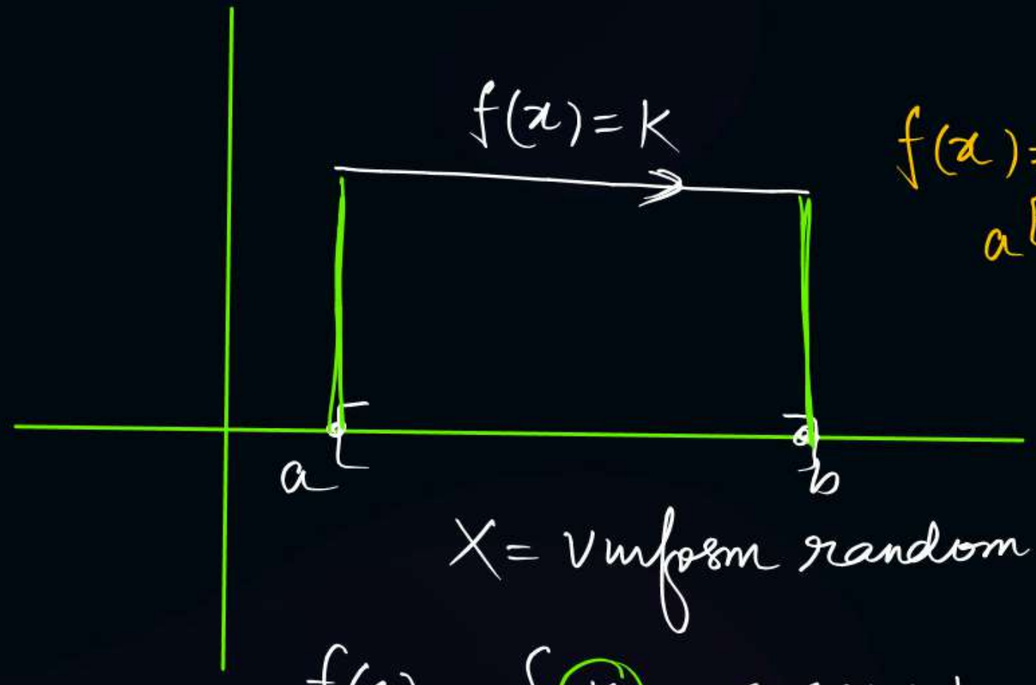
$f(x) = K = \frac{1}{6}$

Uniform continuous Random variable

Prob - (SAME) [0,1]

Function → constant value





$f(x) = K = \frac{1}{(b-a)}$

a [———] b

X = Uniform random variable

$$f(x) = \begin{cases} K & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

U(a, b)

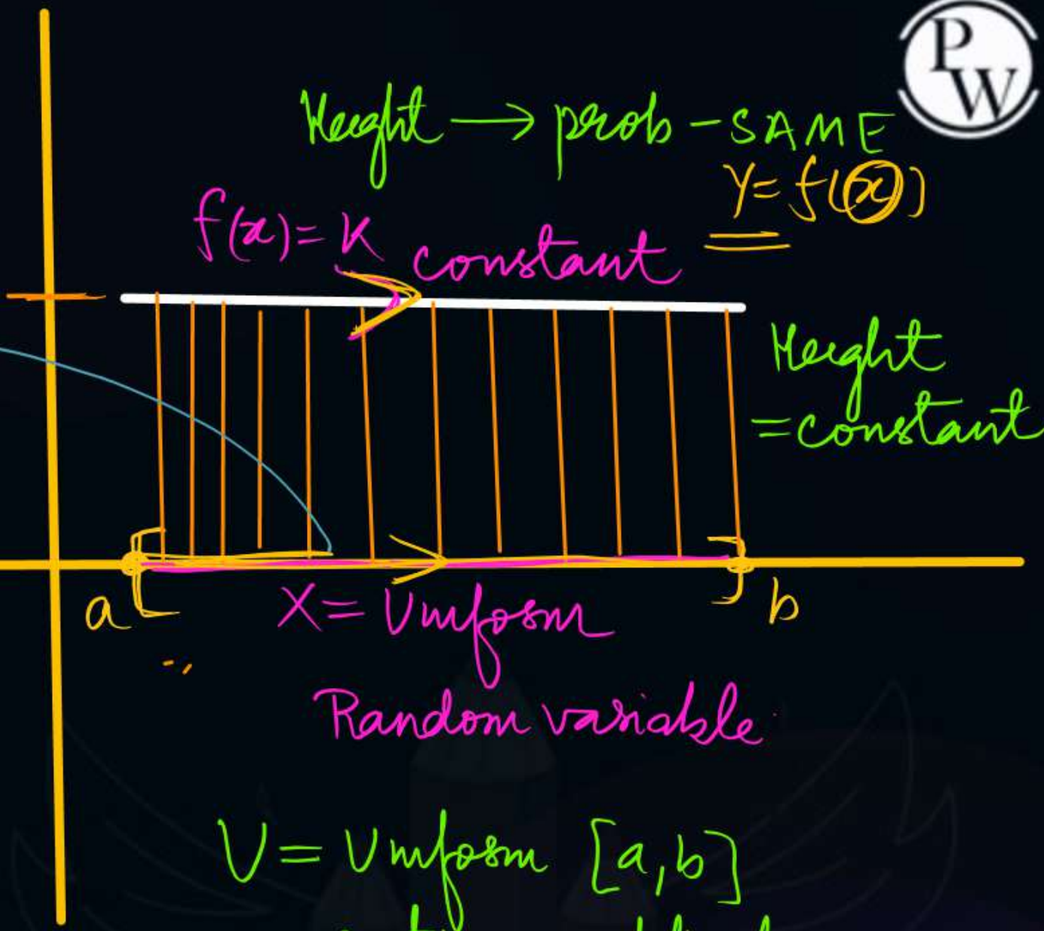
Uniform Random variable [a, b]

Tossing A coin Single

Uniform random

Height → prob - SAME

$Y = f(x)$



Random variable

U = Uniform [a, b]

= continuous defined on Interval

$f(x) = K = \frac{1}{6}$

SAMPLE Points (Infinite

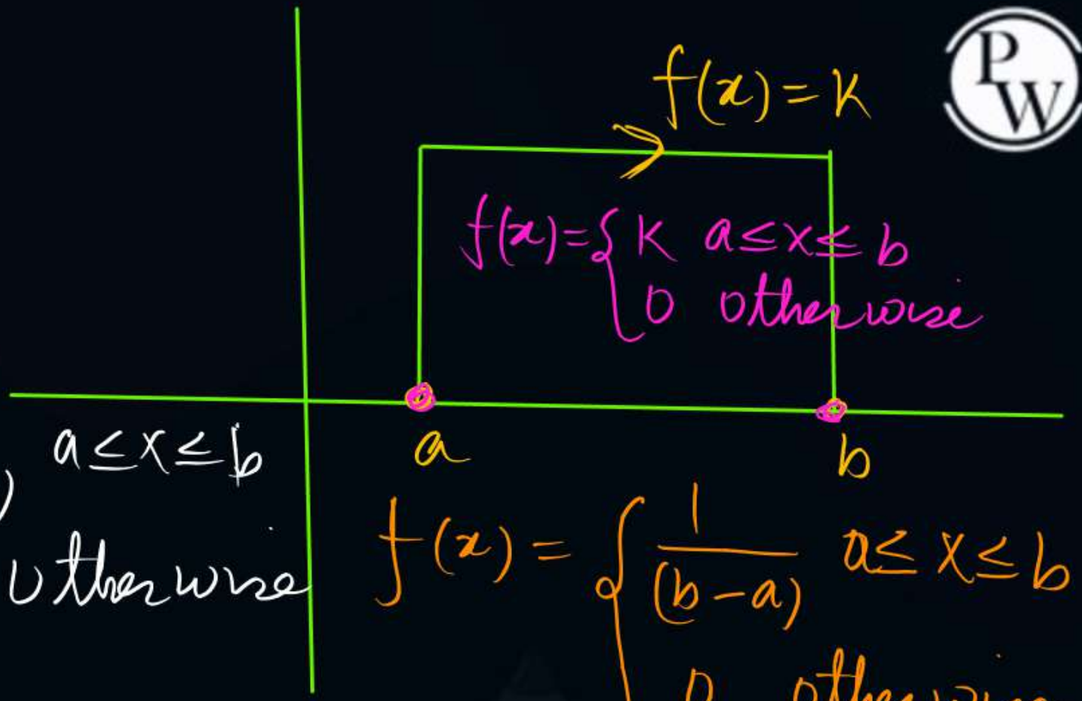
Using valid pdf

$$\int_a^b K\,dx = 1$$

$$= K(b-a) = 1$$

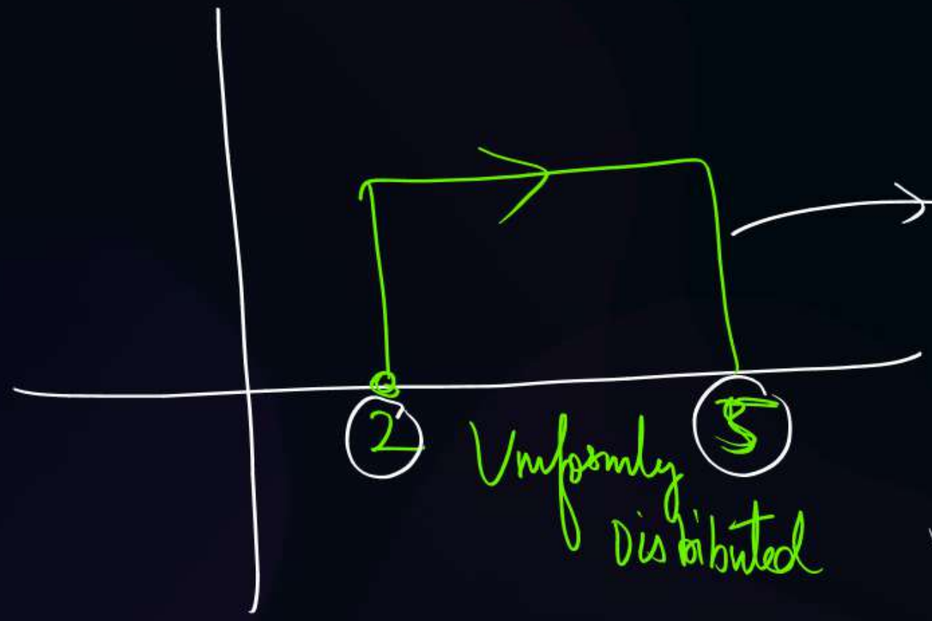
$$K = \frac{1}{(b-a)}$$

$$U(a,b) = \begin{cases} \frac{1}{(b-a)} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$



$$f(x) = K \quad f(x) = \begin{cases} K & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{(b-a)} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$



$$f(x) = \begin{cases} \frac{1}{(5-2)} & 2 \le x \le 5 \text{ (Interval)} \\ 0 & \text{otherwise} \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{3} & 2 \le x \le 5 \\ 0 & \text{otherwise} \end{cases}$$

$$U(a,b) = f(x) = \begin{cases} \frac{1}{(b-a)} & a \le x \le b \\ \text{otherwise } 0 \end{cases}$$

Expected value of Uniform Random variable:

$$E[X] = \int_a^b x f(x)\, dx$$

$$E[X] = \int_a^b x \cdot \frac{1}{(b-a)}\, dx$$

$$= \frac{1}{(b-a)} \int_a^b x\, dx$$

$$= \left[\frac{x^2}{2}\right]_a^b \times \frac{1}{(b-a)}$$

$$= \frac{(b^2-a^2)}{2} \cdot \frac{1}{(b-a)}$$

$$\boxed{E[X] = \frac{(a+b)}{2}}$$

expected value OR mean value

or

Average

Expected value of x.

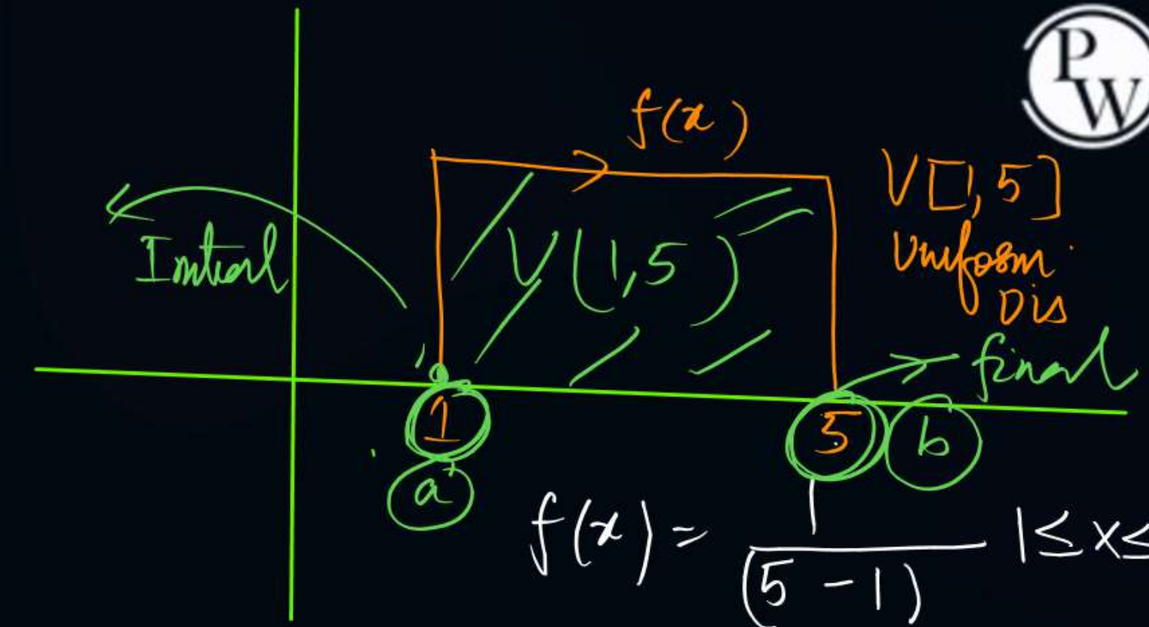
$$E[X] = \int_a^b x f(x)\,dx$$

$$= \int_1^5 x \cdot \frac{1}{4}\,dx$$

$$= \frac{(5^2 - 1^2)}{2} \times \frac{1}{4}$$

$$= \frac{24}{2} \times \frac{1}{4}$$

$$E[X] = 3$$



$$f(x) = \frac{1}{(5-1)} \quad 1 \le x \le 5$$

$$0 \quad \text{otherwise}$$

$$f(x) = \frac{1}{4} \quad 1 \le x \le 5$$

$$= 0 \quad \text{otherwise}$$

$$E[X] = \frac{(a+b)}{2} = \frac{5+1}{2} = 3$$

$$\boxed{Var(X) = E[X^2] - [E[X]]^2}$$

$Var(X) > 0$

$Var(-X) = (-1)^2 Var(X)$

$$E[X^2] = \int_a^b x^2 f(x)\,dx = \int_a^b x^2 \cdot \frac{1}{(b-a)}\,dx$$

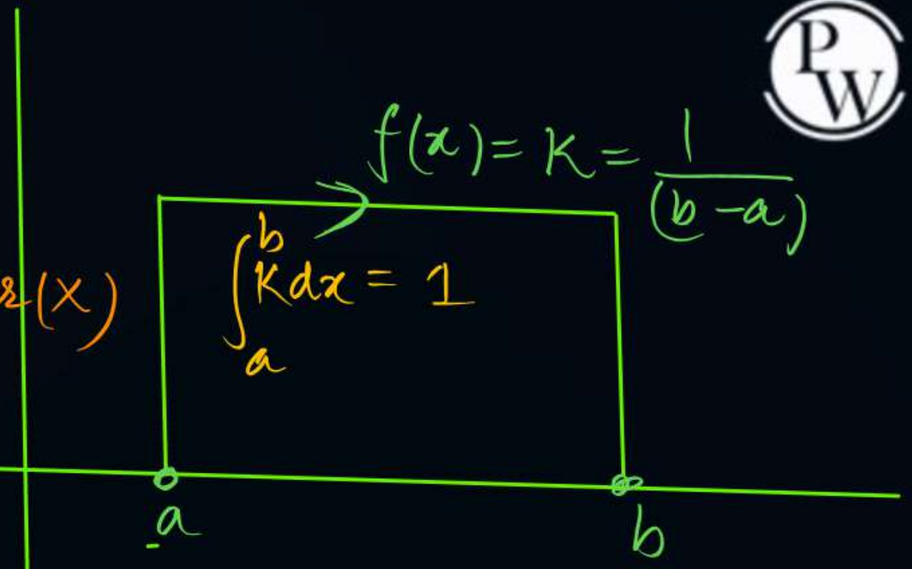
$$= \frac{(b^3-a^3)}{3} \cdot \frac{1}{(b-a)}$$

$$= \frac{(b-a)(b^2+a^2+ab)}{3} \cdot \frac{1}{(b-a)}$$

$$\boxed{E[X^2] = \frac{a^2+b^2+ab}{3}}$$

$$E[X] = \frac{a+b}{2}$$

$$Var(X) = \sigma_X^2 = \frac{(a^2+b^2+ab)}{3} - \left(\frac{a+b}{2}\right)^2$$



Two Parameters $(a, b) \rightarrow V(a,b)$

$$\boxed{Var(X) = \frac{(b-a)^2}{12} \text{ OR } \frac{(a-b)^2}{12}}$$

$$S.D = \sqrt{Var(X)} = \sqrt{\frac{(a-b)^2}{12}}$$



Topic :Concept of gaussian distribution

Normal Distribution : OR Guassian Distribution

If n is Large No of Trials $\xrightarrow{n \to \infty}$ bell shaped

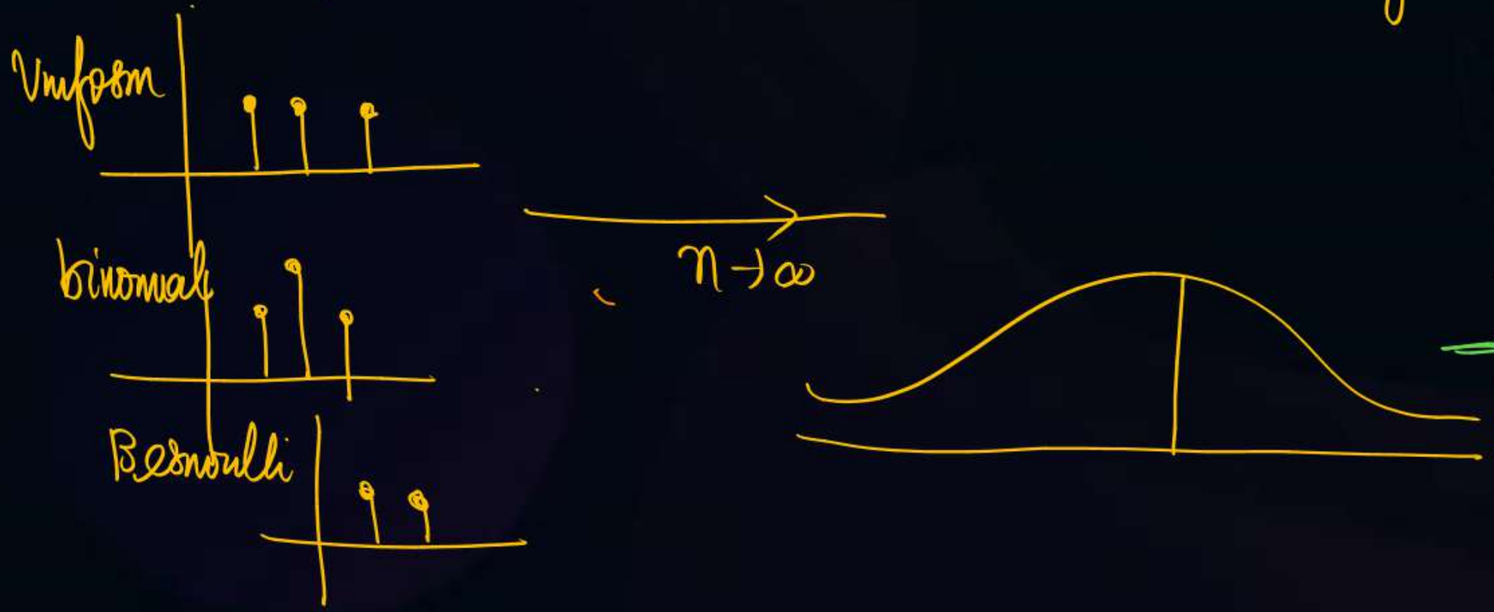


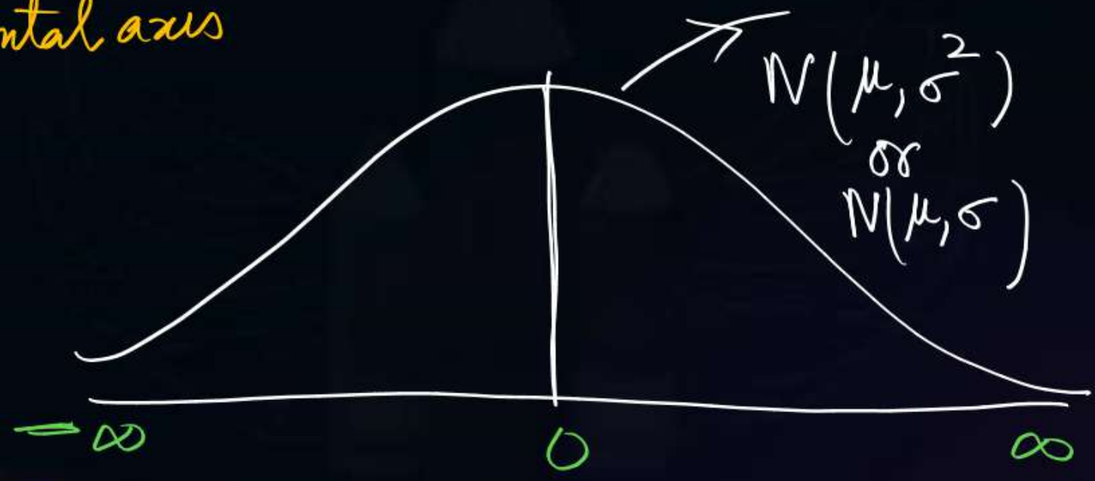
(A) bell shaped curve

(B) This is a continuous Random variable

(C) $N(\mu, \sigma^2)$ = Guassian Distribution

(D) NEVER touches or crosses the Horizontal axis

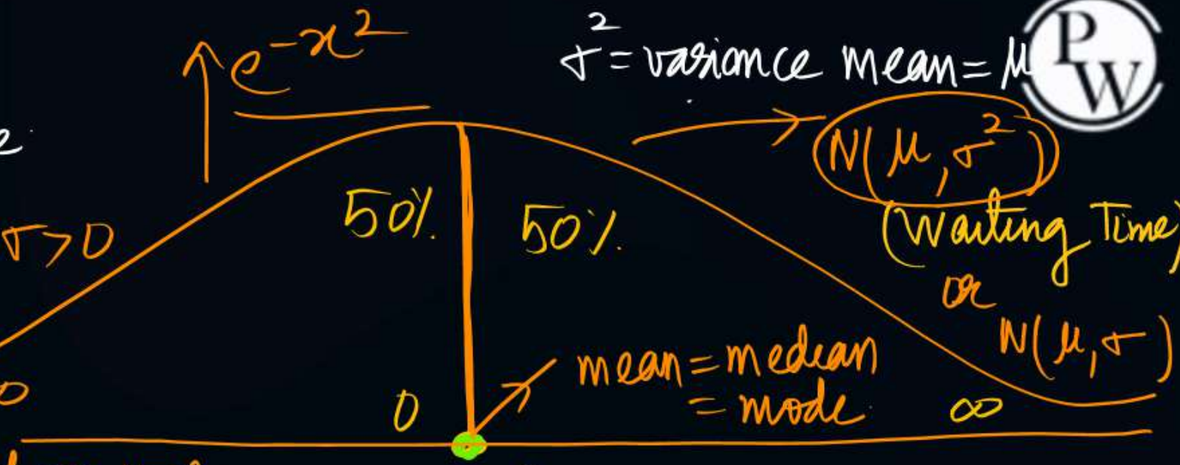




'X' is a Continuous Random variable

$$f(x) = N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \sigma > 0$$

$$-\infty < \mu < \infty$$



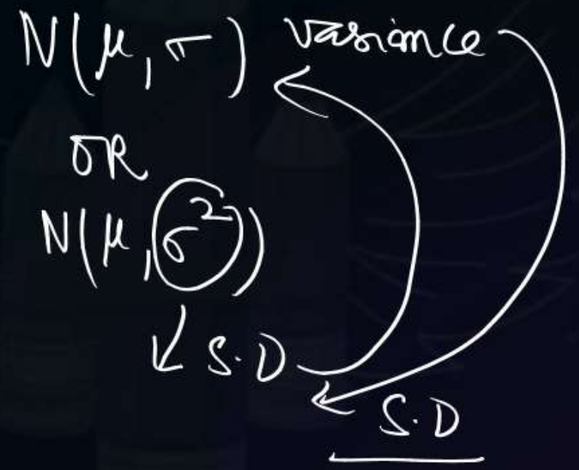
If It is a valid pdf X = random var

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

$$\int_{-\infty}^{\infty} N(\mu, \sigma^2)\, dx = 1$$

✓ symmetric Distribution

✓ mean = median = mode
= coincide



$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

[even Function]

$D = t$

$\infty = t$

$$\frac{x-\mu}{\sigma\sqrt{2}} = t$$

both sides Differentiate It

$$\frac{dx-0}{\sigma\sqrt{2}} = dt \qquad dx = \sigma\sqrt{2}\,dt$$

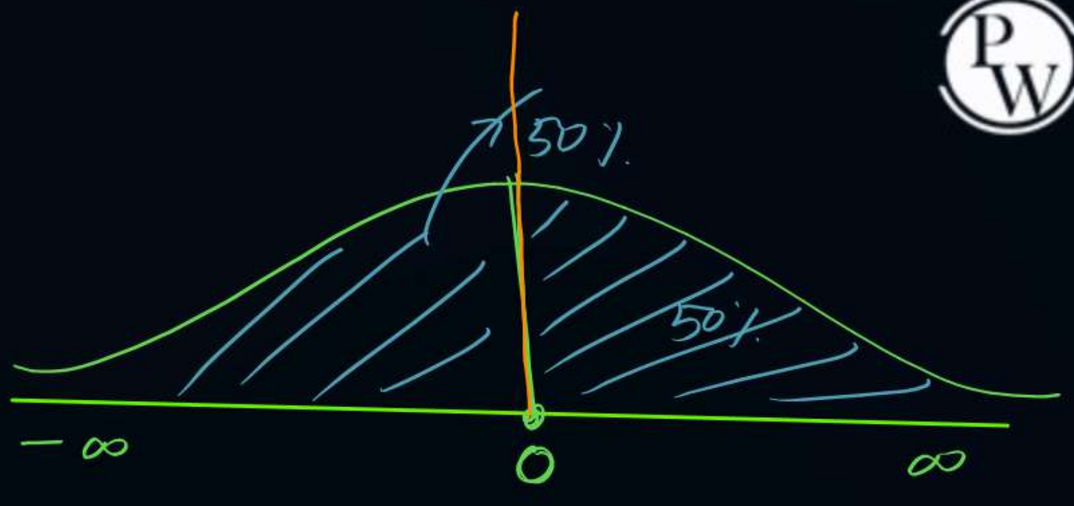
$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-t^2} \cdot \sigma\sqrt{2}\,dt$$

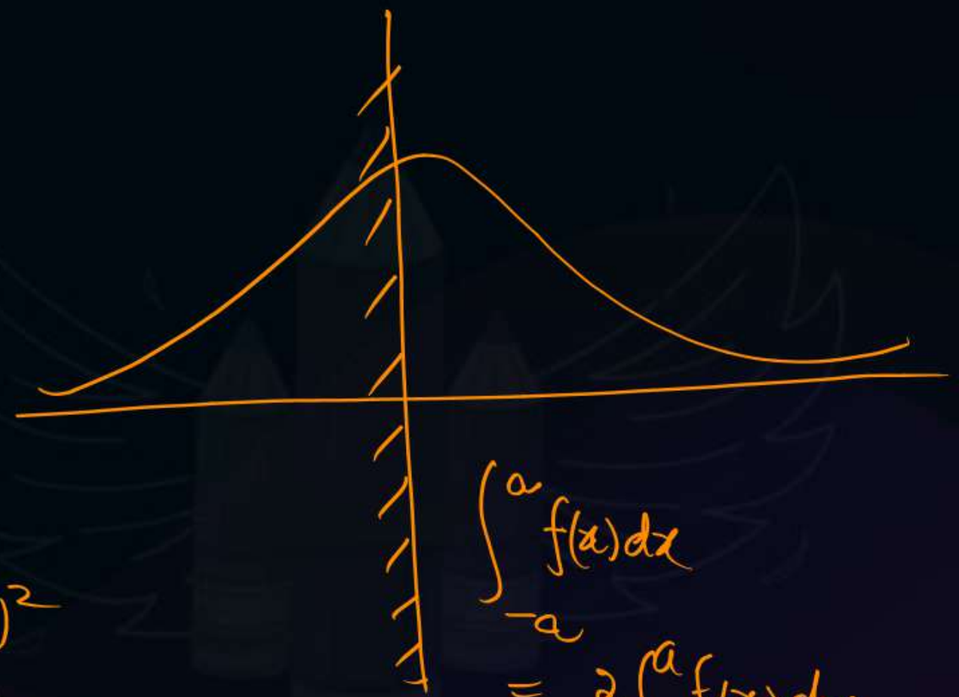
$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt$$ → even Function

$f(-t) = f(t)$

$f(-t) = e^{-(-t)^2}$

$= e^{-t^2}$ even





$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

$$= \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} dt$$

Put $t^2 = z$ $\quad 2t\,dt = dz$

$$dt = \frac{dz}{2t} = \frac{dz}{2\sqrt{z}}$$

$$= \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-z} \cdot \frac{dz}{2\sqrt{z}}$$

$$= \frac{2}{\sqrt{\pi}} \frac{1}{2} \int_0^\infty e^{-z} z^{-1/2} dz$$

$$= \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-z} z^{-1/2} dz$$

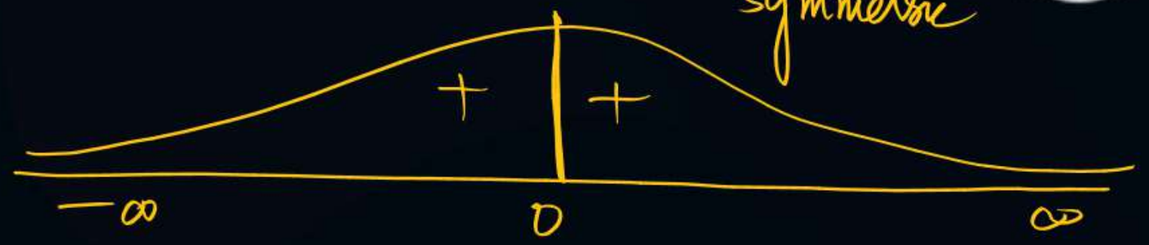
Does Not integrate

Compare

$$z^{-1/2} = z^{n-1}$$

$$n-1 = -\frac{1}{2}$$

$$n = \frac{1}{2}$$



Γn = gamma Function

$$\Gamma n = \int_0^\infty e^{-z} z^{n-1} dz$$

$\Gamma n = (n-1)!$

$\Gamma \frac{1}{2} = \sqrt{\pi}$

$\Gamma 5 = 4!$ $\quad \Gamma 6 = 5!$

$\Gamma 3 = 2!$

$$= \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-z} z^{-1/2} dz$$

$$= \frac{1}{\sqrt{\pi}} \times \Gamma\frac{1}{2} = \frac{\sqrt{\pi}}{\sqrt{\pi}} \times 1 = 1$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

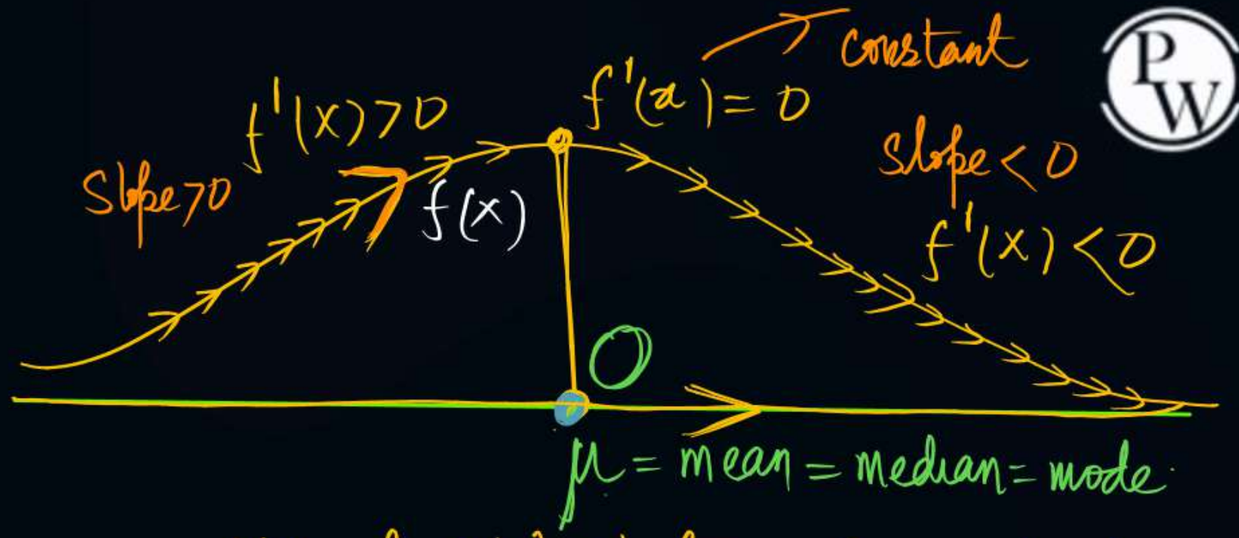
$$f'(x) = 0$$

$$\frac{d}{dx}\left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right] = 0$$

$$\boxed{x = \mu}$$

$$\frac{d^2y}{dx^2} = 0$$

$$-\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\left[1 - \left(\frac{x-\mu}{\sigma^2}\right)\right] = 0$$



Normal Distribution Sym. about mean



- ✓ Point of Inflection
- ✓ Point of contraflexure
- ✓ Sudden change

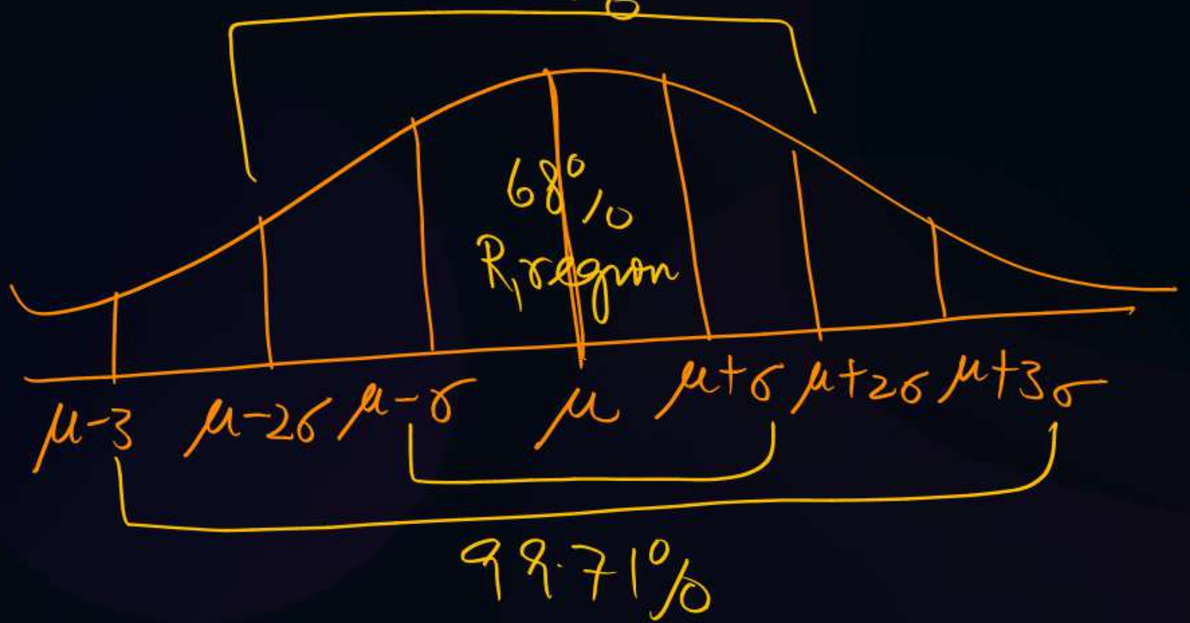
$$= -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \left[1 - \frac{(x-\mu)^2}{\sigma^2}\right] = 0$$

$$\Rightarrow \quad 1 - \frac{(x-\mu)^2}{\sigma^2} = 0$$

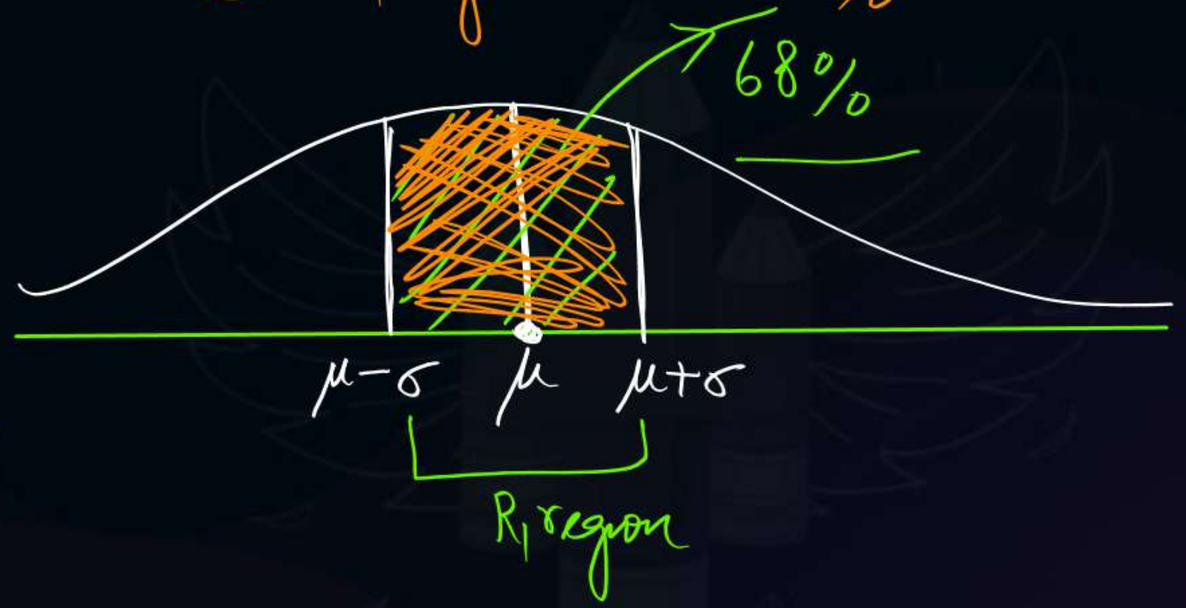
$$\Rightarrow \quad \boxed{x = \mu \pm \sigma}$$

$x = \mu \pm 2\sigma$
$x = \mu \pm 3\sigma$

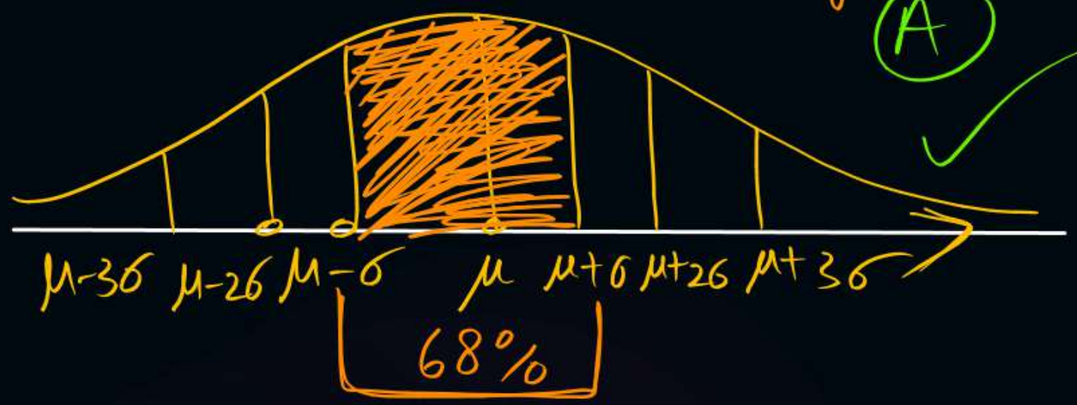
95%



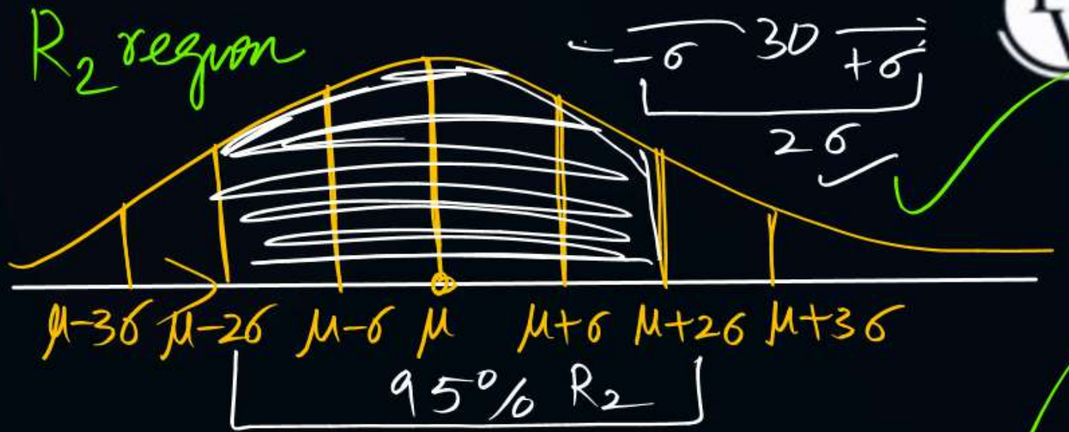
(A) R_1 region = 68%



R_1 region

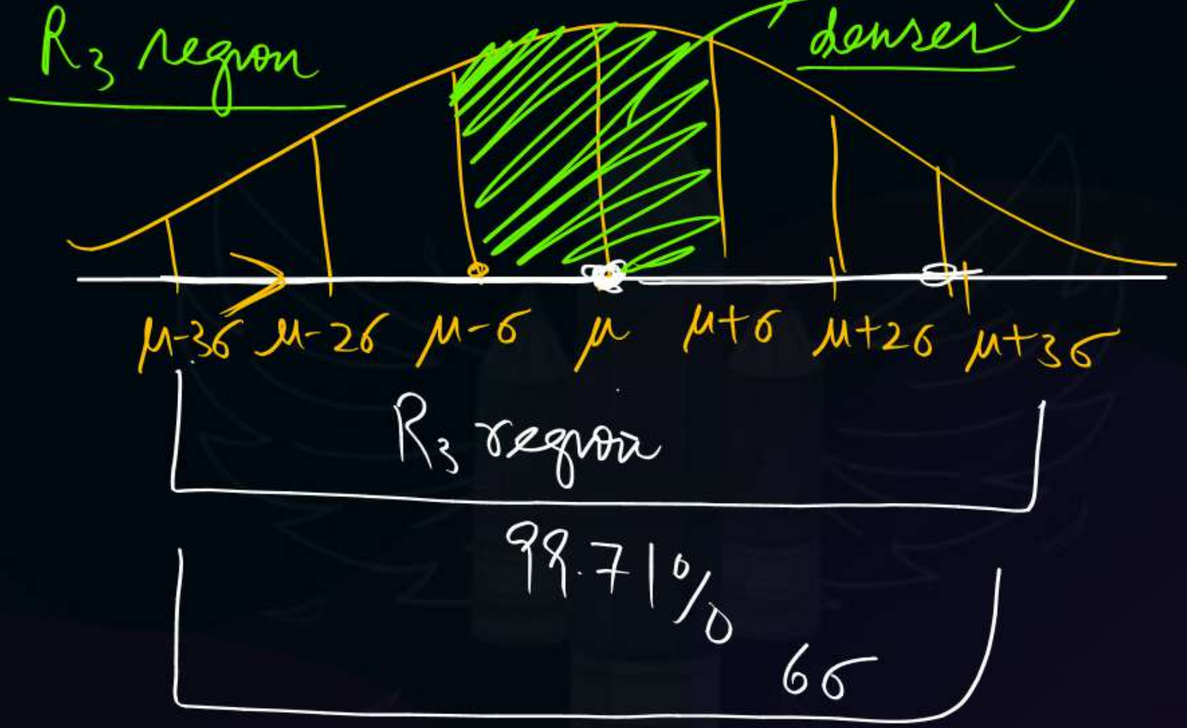


R_2 region



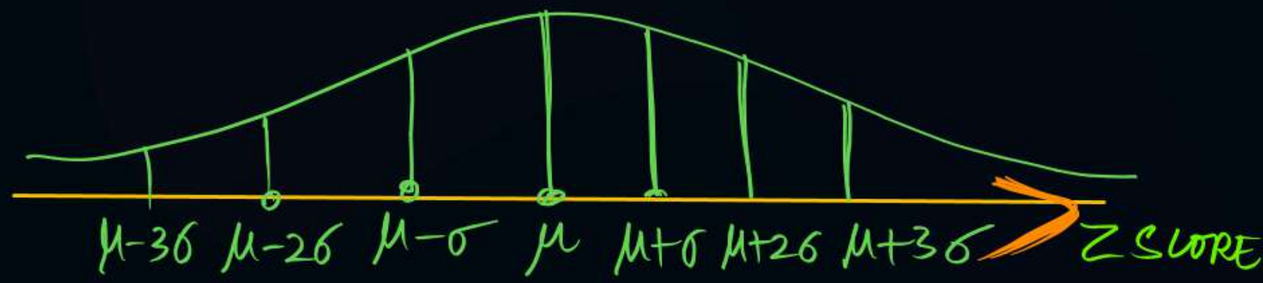
R_3 region

denser



(Normal Distribution Curves)

$Z = \frac{x-\mu}{\sigma}$ = statistical Parameter



" Marks Distribution

Tamilnadu → μ, σ
Maharastra → μ, σ
Karnataka → μ, σ
Goa →
Kerala
U.P
bihar
guzrat
M.P

μ, σ vary

Standasized Z = statistical Parameter

$$Z = \frac{x-\mu}{\sigma}$$

marks - Stanadize

- ✓ Uniform + guassian
- ✓ Integration Basics : GATE wallah
- ✓ DPP.

THANK - YOU